

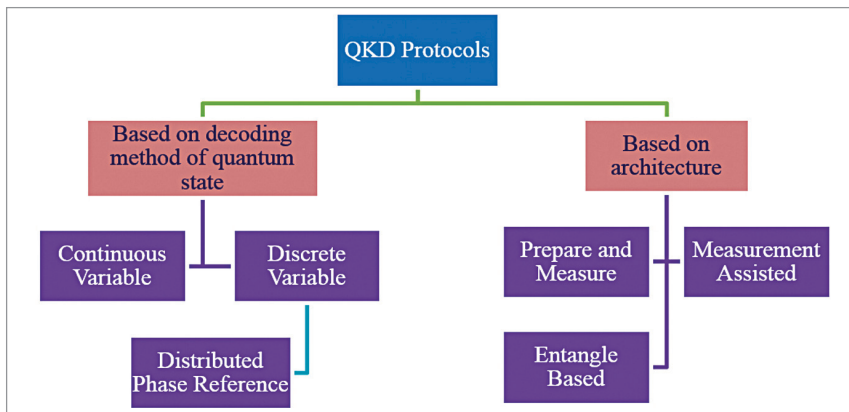


Fig. 3: Types of QKD protocols

that is pre-authenticated. It can be any IP/Ethernet link.

- Alice sends the quantum information, called qubits, over the quantum channel. Depending upon the QKD protocol used Alice can use the quantum properties of light, such as polarisation, phase, or time-of-arrival.

- Bob performs quantum measurements on the received qubits as per the QKD protocol used. There are always some losses in the quantum channel.

- Alice and Bob share some information over the classical channel to arrive at a common secret key. Eavesdropping on the classical channel does not affect the secrecy of the key.

- The raw key undergoes a sifting process in which Alice and Bob discard incompatible qubits based on the protocol used.

- Alice and Bob check the quantum bit error rate (QBER) to check for eavesdropping (by an adversary, commonly called Eve) on the quantum channel. If QBER is less than a defined threshold, the key is further processed. Otherwise, the communication is aborted.

- Due to imperfections in the source, detector, and transmission media, the sifted keys may have some errors that are corrected using error correction algorithms.

- Finally, the keys undergo a privacy amplification process. The keys

are compressed by suitable factors to conceal any information that may have gone to Eve during the key generation process.

- The generated keys are stored along with key-ID and delivered to the application on request.
- Keys generated in the QKD system can be mathematically proved to be secure and it is not based on any assumption.

QKD protocols

Charles Bennett and Gilles Brassard proposed the first quantum cryptography protocol in 1984 at a conference in Bangalore (now Bengaluru). This protocol is popularly known as BB84. After that several QKD protocols have been proposed and proved to be information-theoretically secure. The evolution of various QKD protocols was influenced by the following factors:

- Ease of implementation on a physical system
- Rate of secret key generation
- Supported distance between Alice and Bob
- Immunity from quantum hacking activities that exploit the imperfections in QKD devices

Based on the decoding method of quantum states, QKD protocols can be broadly divided into two categories:

Continuous variable (CV) QKD.

Here information is encoded using the position and momentum quad-

rature of a quantised electromagnetic field in an infinite-dimensional Hilbert space.

Discrete variable (DV) QKD.

DV-QKD protocols encode information with discrete variables of finite dimension, such as phase, polarisation, or time bin of single photons.

A special type of DV-QKD protocol uses the time and phase of weak coherent pulses to prepare qubits, and coherence between consecutive pulses are checked to identify eavesdropping. This is called distributed phase reference (DPR) protocol.

Differential phase shift (DPS) and coherent one-way (COW) are examples of DPR protocols. They are widely used due to the long range and high rate of key generation.

On the other side, based on the architecture of protocols, QKD protocols can be grouped into three categories:

Prepare and measure (P&M).

A transmitter (Alice) prepares and sends a quantum state to a receiver (Bob) that measures it to arrive at the final shared key. Example: BB84 protocol.

Measurement assisted. Two transmitters prepare and send quantum states to a receiver (traditionally called Charlie) that performs a joint measurement. The final key is shared between two transmitters. Here both Alice and Bob act as transmitters. Examples: Measurement Device-Independent (MDI) and Twin-Field (TF) QKD protocols.

Entangle based. A transmitter (Charlie) prepares a bipartite entangled quantum state and sends its two parts to two receivers that measure the quantum state. The final key is shared between two receivers. Here both Alice and Bob act as receivers. Example: E91 QKD protocol.

Standardisation activities

Different standard development organisations—like the European Telecommunication Standards Institute (ETSI), the International Telecom-